

Problem Set 2, COSC330/PHYS306 Sections 2-8, 2-11, and 3-2 through 3-6

Prof. Jordan C. Hanson

September 11, 2026

1 Chapter 2

1. [Section 2-8, Exercise 38] Convert each binary number to hexadecimal:

- (a) $1110 = E$
- (b) $10 = 2$
- (c) $10111 = 17$
- (d) $10100110 = A_{16}$
- (e) $1111110000 = 3F0$
- (f) $100110000010 = 982$

2. [Section 2-8, Exercise 39] Convert each hexadecimal number to decimal:

- (a) $23_{16} = (2 \cdot 16^1) + (3 \cdot 16^0) = 32 + 3 = 35$
- (b) $92_{16} = (9 \cdot 16^1) + (2 \cdot 16^0) = 144 + 2 = 146$
- (c) $1A_{16} = (1 \cdot 16^1) + (10 \cdot 16^0) = 16 + 10 = 26$
- (d) $8D_{16} = (8 \cdot 16^1) + (13 \cdot 16^0) = 128 + 13 = 141$
- (e) $F3_{16} = (14 \cdot 16^1) + (11 \cdot 16^0) = 224 + 11 = 235$
- (f) $EB_{16} = (14 \cdot 16^1) + (11 \cdot 16^0) = 224 + 11 = 235$
- (g) $5C2_{16} = (5 \cdot 16^2) + (12 \cdot 16^1) + (2 \cdot 16^0) = 1280 + 192 + 2 = 1474$
- (h) $700_{16} = (7 \cdot 16^2) + (0 \cdot 16^1) + (0 \cdot 16^0) = 1792 + 0 + 0 = 1792$

3. [Section 2-11, Exercise 56] Convert each binary number to Gray code:

- (a) $11011 = 10110$
- (b) $1001010 = 1101111$
- (c) $1111011101110 = 1000110011001$

4. [Section 2-11, Exercise 57] Convert each Gray code to binary:

- (a) $1010 = 1100$
- (b) $00010 = 00011$
- (c) $11000010001 = 10000011110$

2 Chapter 3

5. [Section 3-2, Exercise 6] Repeat Problem 5 for the waveforms in Figure 3-79.

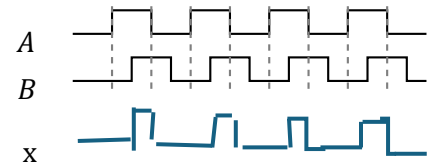


FIGURE 3-79

6. [Section 3-2, Exercise 7] The input waveforms applied to a 3-input AND gate are as indicated in Figure 3-80. Show the output waveform in proper relation to the inputs with a timing diagram.

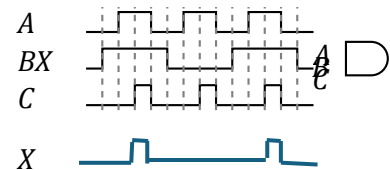


FIGURE 3-80

7. [Section 3-3, Exercise 12] Repeat Problem 8 for a 4-input OR gate.

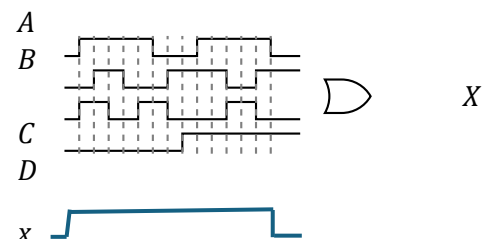
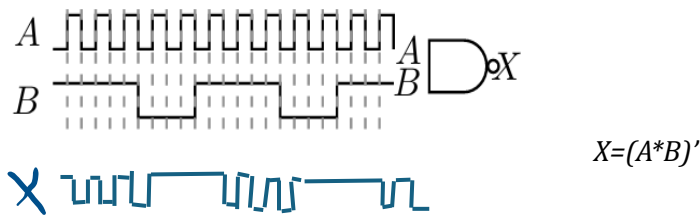


FIGURE 3-81

[Section 3-4, Exercise 17] For the set of input waveforms in Figure 3-83, determine the output for the gate shown and draw the timing diagram.



9. **[Section 3-4, Exercise 18]** Determine the gate output for the input waveforms in Figure 3-84 and draw the timing diagram

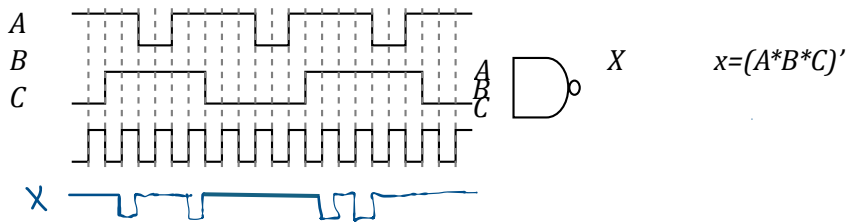
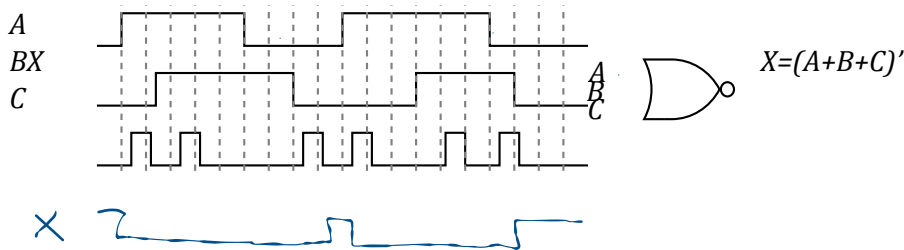
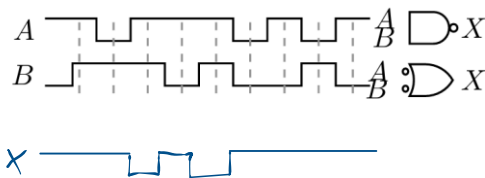


FIGURE 3-84

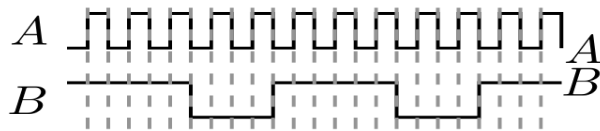
10. **[Section 3-5, Exercise 22]** Determine the output waveform in Figure 3-87 and draw the timing diagram.



Section 3-5, Exercise 24] The NAND and then the negative OR symbols represent equivalent operations, but they are functionally different. For the NOR symbol, look for at least one HIGH on the inputs to give a LOW on the output. For the negative-AND, look for two LOWs on the inputs to give a HIGH output. Using these two functional points of view, show that both gates in Figure 3-88 will produce the same output for the given inputs.



12. **[Section 3-6, Exercise 26]** Repeat Problem 17 for an exclusive-OR gate.



13. **[Section 3-6, Exercise 27]** Repeat Problem 17 for an exclusive-NOR gate.

